

Unit 4.5. Curve Sketching

Note Title

2012/05/06

Stewart 4.5 p310 - p318

and also read examples

6, 7, 8 on p 295 - 296.

Points on how to sketch
a function in Study Guide

There are two examples
on clickup

(A polynomial & rational
function). Do!

To sketch a function use the following points:

1. Find the domain of the function.
2. Find the intercepts of the function with the x -axis and y -axis.
3. Determine whether the function is even, odd, or periodic, or none of these.

4. Find The horizontal asymptotes (if any) of the function.

5. Find The vertical asymptotes (if any) of the function.

6. Find interval(s) on which the function is increasing / decreasing.

7. Find local extremes.

8. Find interval(s) on which the function is concave up and concave down.

9. Find inflection points of the function

10. use the information to sketch the graph of the function.

Examples: Sketch

1. $f(x) = x e^{-x}$

2. $f(x) = \frac{e^{2x}}{x+1}$

3

$$f(x) = \ln(1+x^2)$$

4

$$f(x) = (1 - \ln x)^2$$

Example 1

Note Title

2013/05/03

$$f(x) = x e^{-x}$$

1) Domain = $\mathbb{R}$

2) x-int and y-int

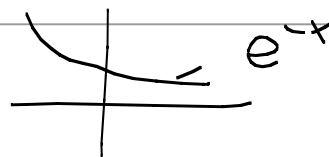
$$f(0) = 0$$

3) $f(-x) = -x e^{-(-x)} = -x e^x$ Not even or odd

4) $\lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} x e^{-x}$ Form $(\infty \cdot 0)$

$$= \lim_{x \rightarrow \infty} \frac{x}{e^x} \quad \text{Form } \left(\frac{\infty}{\infty} \right)$$

$$\stackrel{\text{d'ltop}}{=} \lim_{x \rightarrow \infty} \frac{1}{e^x} = 0$$



Note Title

2013/05/03

$$\lim_{x \rightarrow -\infty} x e^{-x} = -\infty$$

$\therefore y = 0$ is a H.A.

5) No V.A. because the domain is $\mathbb{R}$.

$$6) f'(x) = e^{-x} - x e^{-x} = e^{-x} [1 - x]$$

$$f'(x) = 0 \Rightarrow 1 - x = 0 \Rightarrow x = 1$$

$$\begin{array}{c} f'(x) > 0 & f'(x) < 0 \\ \hline & 1 \end{array}$$

f is increasing on $(-\infty, 1)$ and decreasing on $(1, \infty)$

7) At $x=1$ The function has a local max
of $f(1) = e^{-1}$

$$\begin{aligned} 8) f''(x) &= -e^{-x} - [e^{-x} - xe^{-x}] = -e^{-x} - e^{-x} + xe^{-x} \\ &= e^{-x} [-2 + x] \end{aligned}$$

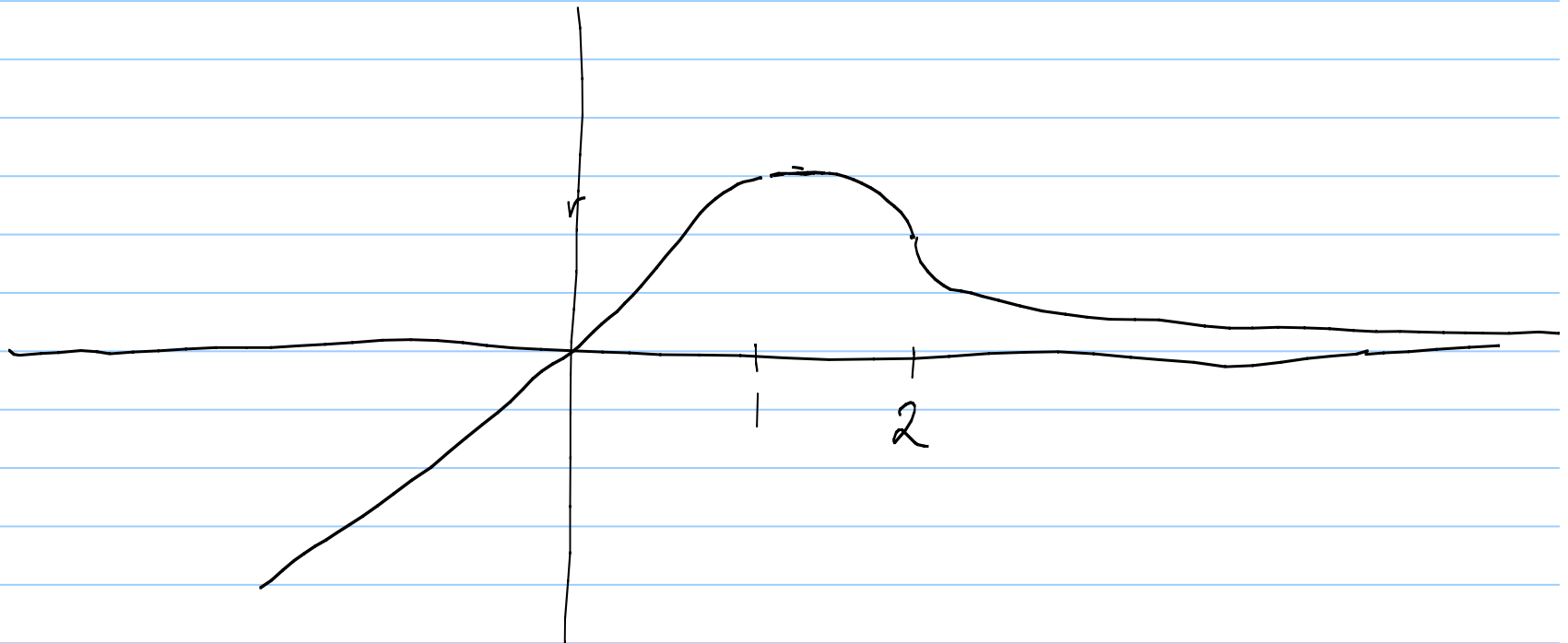
$$f''(x) = 0 \Rightarrow -2 + x = 0 \Rightarrow x = 2$$

$$\begin{array}{c} f'' < 0 \qquad \qquad f'' > 0 \\ \hline \qquad \qquad \qquad 2 \end{array}$$

That the function is concave down on $(-\infty, 2)$ and
the function is concave up on $(2, \infty)$

9) The function has a inflection point at $(2, 2e^{-2})$

10)



Example 2

Note Title

2013/05/03

$$f(x) = \frac{e^{2x}}{x+1}$$

1) Domain = $\mathbb{R} \setminus \{-1\}$

2) No x -int and we have a y -int let $x=0$

$$f(0) = \frac{e^0}{1} = e^0 = 1$$

at $y=1$.

3) $f(-x) = \frac{e^{2(-x)}}{-x+1}$, Not even or odd.

$$4) \lim_{x \rightarrow \infty} \frac{e^{2x}}{x+1} \quad \text{Form } \left(\frac{\infty}{\infty} \right)$$

$$\stackrel{\text{L'Hop}}{=} \lim_{x \rightarrow \infty} \frac{2e^{2x}}{1} = \infty$$

$$\lim_{x \rightarrow -\infty} \frac{e^{2x}}{x+1} = 0$$

$\therefore y=0$ is a H.A.

$$5) \lim_{x \rightarrow -1^-} \frac{e^{2x}}{x+1} = -\infty$$

$$\lim_{x \rightarrow -1^+} \frac{e^{2x}}{x+1} = \infty$$

∴ $x = -1$ is a V.A.

$$6) f'(x) = \frac{2e^{2x}(x+1) - e^{2x} \cdot 1}{(x+1)^2}$$

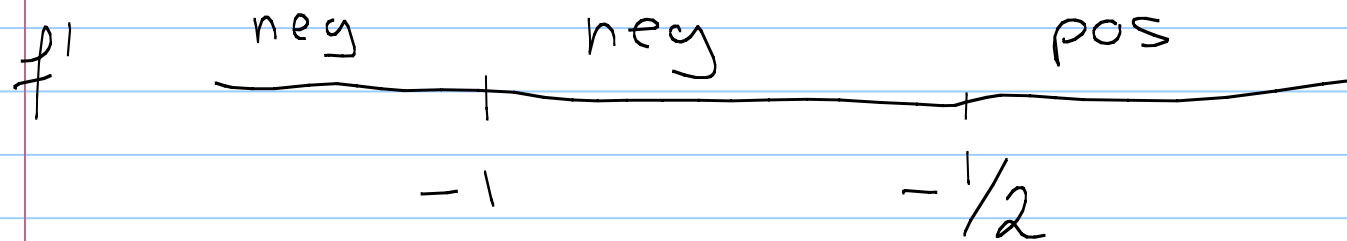
$$f'(x) = 0 \Rightarrow e^{2x} [2x + 2 - 1] = 0$$

$$\Rightarrow 2x + 1 = 0$$

$$\Rightarrow x = -\frac{1}{2}$$

at $x = -1$ $f'(x)$ does not exist

critical points at $x = -1$ and $x = -\frac{1}{2}$



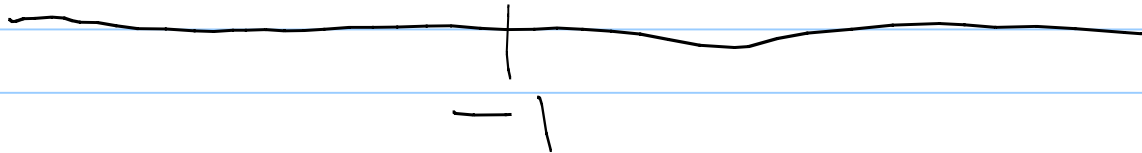
The function is decreasing on $(-\infty, -1)$ and on $(-1, -\frac{1}{2})$ and increasing on $(-\frac{1}{2}, \infty)$

7) at $x = -\frac{1}{2}$ The function has a local min
of $f(-\frac{1}{2}) = e^{2(-\frac{1}{2})} / (\frac{1}{2} + 1) = 2e^{-1}$

$$f''(x) = \frac{2e^{2x}(2x+1)(x+1)^2 + 2e^{2x}(x+1)^2 - 2e^{2x}(2x+1)(x+1)}{(x+1)^4}$$

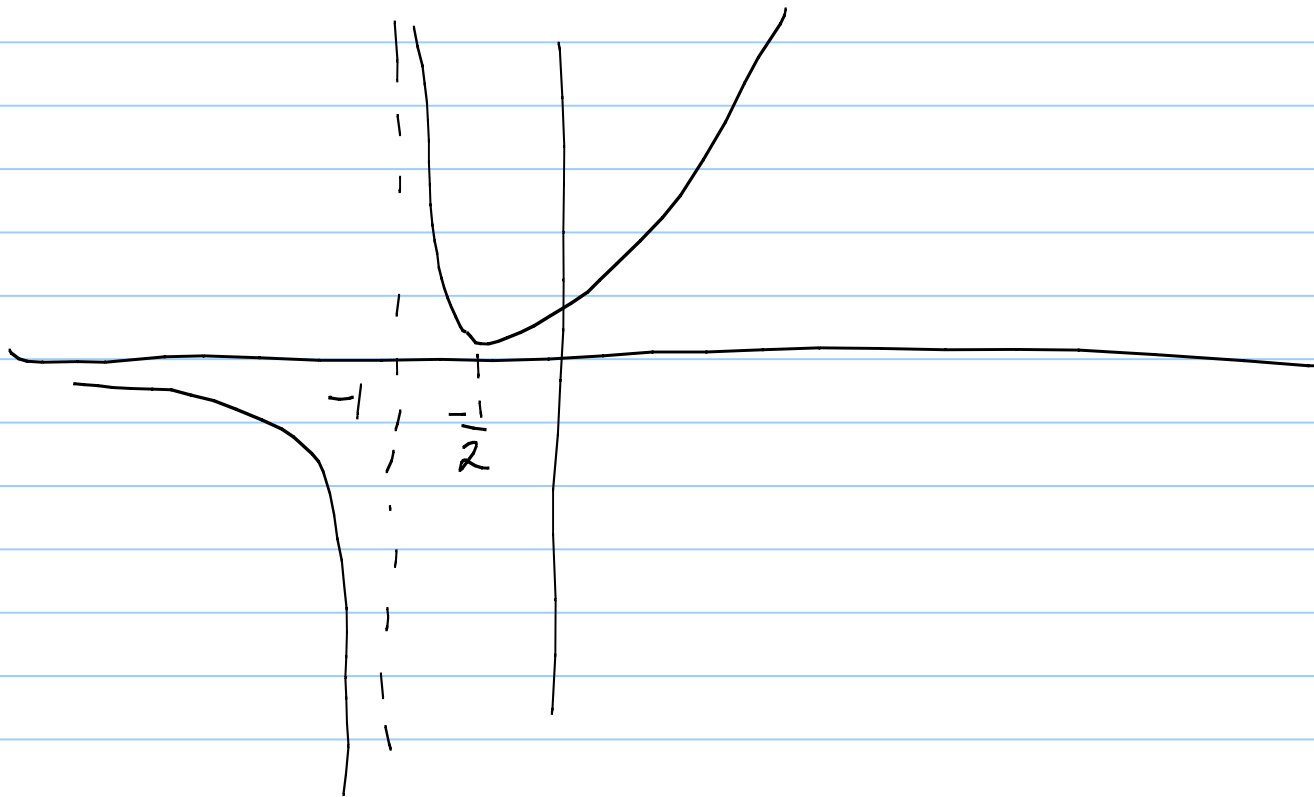
$$= \frac{2e^{2x}(2x^2 + 2x + 1)}{(x+1)^3}$$

$f''(x)$ does not exist at $x = -1$



The function f is concave down on $(-\infty, -1)$ and concave up on $(-1, \infty)$

9) There is no inflection point.



Example 3

$$f(x) = \ln(1+x^2)$$

1) Domain : $\{x \mid x \in \mathbb{R}\}$

2) $f(0) = 0$

3) $f(-x) = \ln(1+(-x)^2) = \ln(1+x^2) = f(x)$

$\therefore$ even

4) $\lim_{x \rightarrow \infty} \ln(x^2+1) = \infty$

$$\lim_{x \rightarrow -\infty} \ln(x^2 + 1) = \infty$$

∴ No horizontal asym.

5) No Vertical asym.

$$6) f'(x) = \frac{1}{x^2 + 1} \cdot 2x$$

$$\frac{2x}{x^2 + 1} = 0 \Rightarrow x = 0$$

$$\begin{array}{ccc} f' < 0 & & f' > 0 \\ \hline & 0 & \end{array}$$

$\therefore f$ is increasing on $(0, \infty)$
 f is decreasing on $(-\infty, 0)$

7) Local extremes:

At $x=0$ The function has a
local min at $f(0)=0$

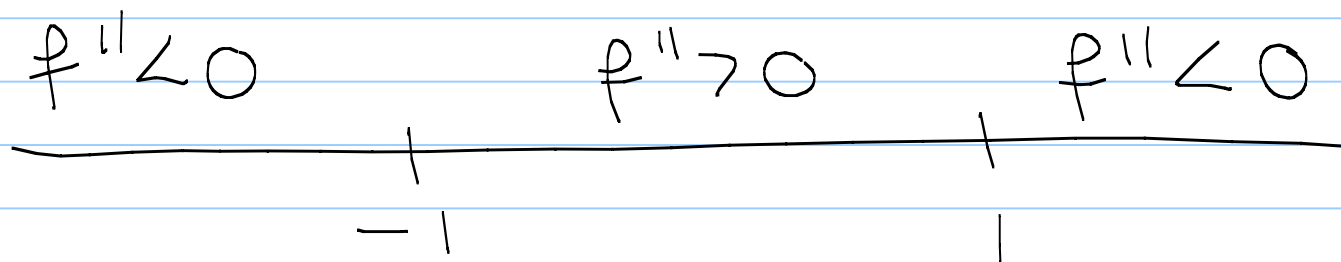
$$8) f''(x) = \frac{2(x^2+1) - 2x \cdot 2x}{(x^2+1)^2}$$

$$\Rightarrow \frac{2x^2 + 2 - 4x^2}{(x^2 + 1)^2} = 0$$

$$\Rightarrow -2x^2 + 2 = 0$$

$$\Rightarrow 1 - x^2 = 0$$

$$\Rightarrow x = \pm 1$$



f is concave down on $(-\infty, -1)$
and $(1, \infty)$

f is concave up on $(-1, 1)$

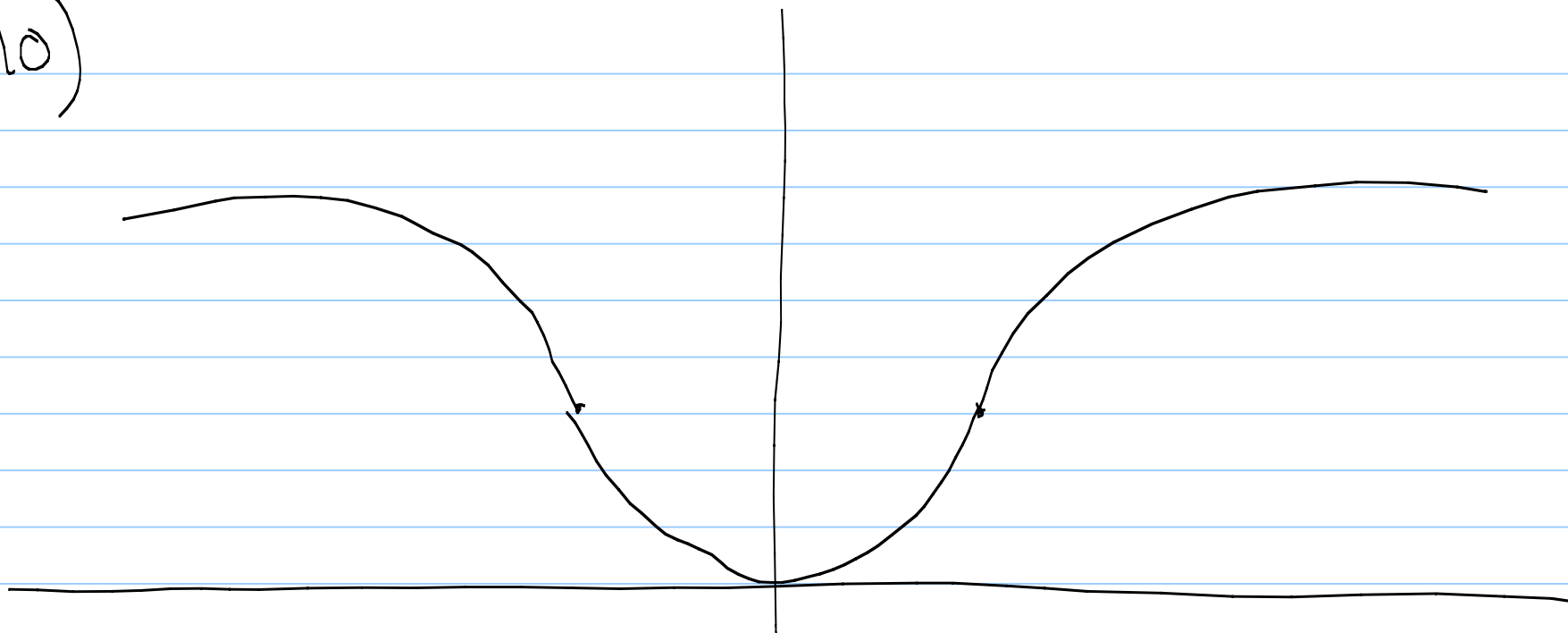
a) Inflection points are
at $x = -1$ with function value

$$f(-1) = \ln(1+1) = \ln 2$$

and at $x = 1$ with

$$f(1) = \ln(1+1) = \ln 2$$

10)



Example 4

$$f(x) = (1 - \ln x)^2$$

1) Domain: $D_f = \{x \mid x > 0\}$

2) No y-int

for x-int let $y = 0$

$$0 = (1 - \ln x)^2$$

$$\Rightarrow 1 - \ln x = 0$$

$$\Rightarrow \ln x = 1$$

$$\Rightarrow x = e$$

$$3) f(-\infty) = (1 - \ln(-\infty))^2 \quad \therefore \text{Neither}$$

$$4) \lim_{x \rightarrow \infty} (1 - \ln x)^2 = \infty$$

Not $H_0 A_0$.

$$5) \lim_{x \rightarrow 0^+} (1 - \ln x)^2 = \infty$$

$\therefore x=0$ is a $V_0 A_0$.

6) Increasing / decreasing

$$f'(x) = 2(1 - \ln x) \cdot \left(-\frac{1}{x}\right)$$

For a critical number

$$f'(x) = 0$$

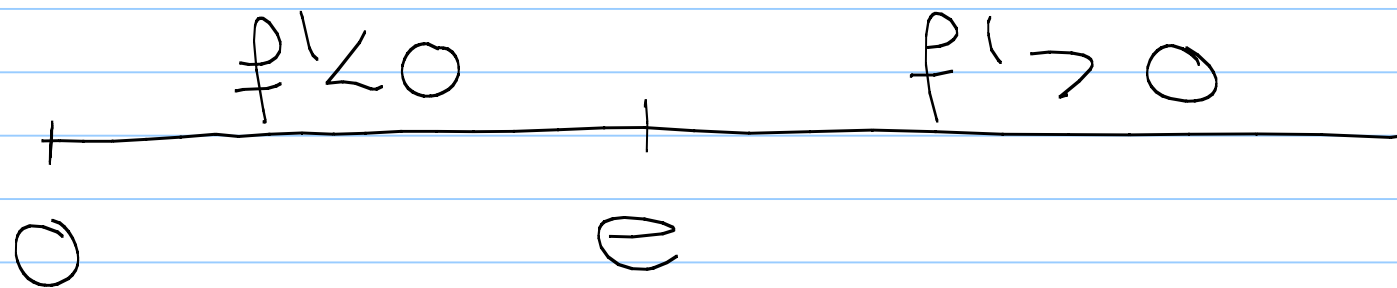
$$\Rightarrow 2(1 - \ln x) \left(-\frac{1}{x}\right) = 0$$

$$\Rightarrow 1 - \ln x = 0$$

$$\Rightarrow \ln x = 1$$

$$\Rightarrow x = e$$

$c = e$ is a critical number.



f is increasing on (e, ∞)
and f is decreasing on
 $(0, e)$

7) Local extremes

There is a local min at

$$f(e) = (1 - \ln e)^2 = 0$$

$$\begin{aligned} 8) \quad f'(x) &= -2(1 - \ln x)\left(\frac{1}{x}\right) \\ &= -\frac{2}{x} + 2 \ln x \left(\frac{1}{x}\right) \end{aligned}$$

$$f''(x) = \frac{2}{x^2} + \frac{2}{x^2} + 2 \ln x \left(-\frac{1}{x^2} \right)$$

$$= \frac{2}{x^2} (1 + 1 - \ln x)$$

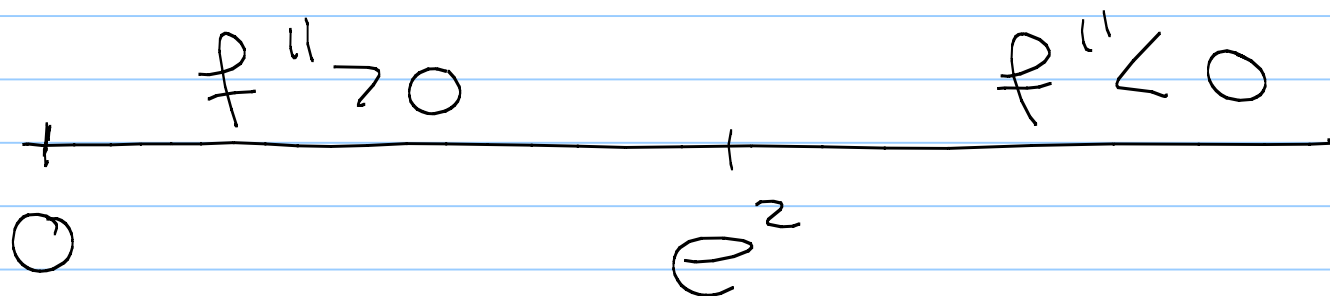
$$= \frac{2}{x^2} (2 - \ln x)$$

$$f''(x) = 0$$

$$\Rightarrow \frac{2}{x^2} (2 - \ln x) = 0$$

$$\Rightarrow \ln x = 2$$

$$\Rightarrow x = e^2$$

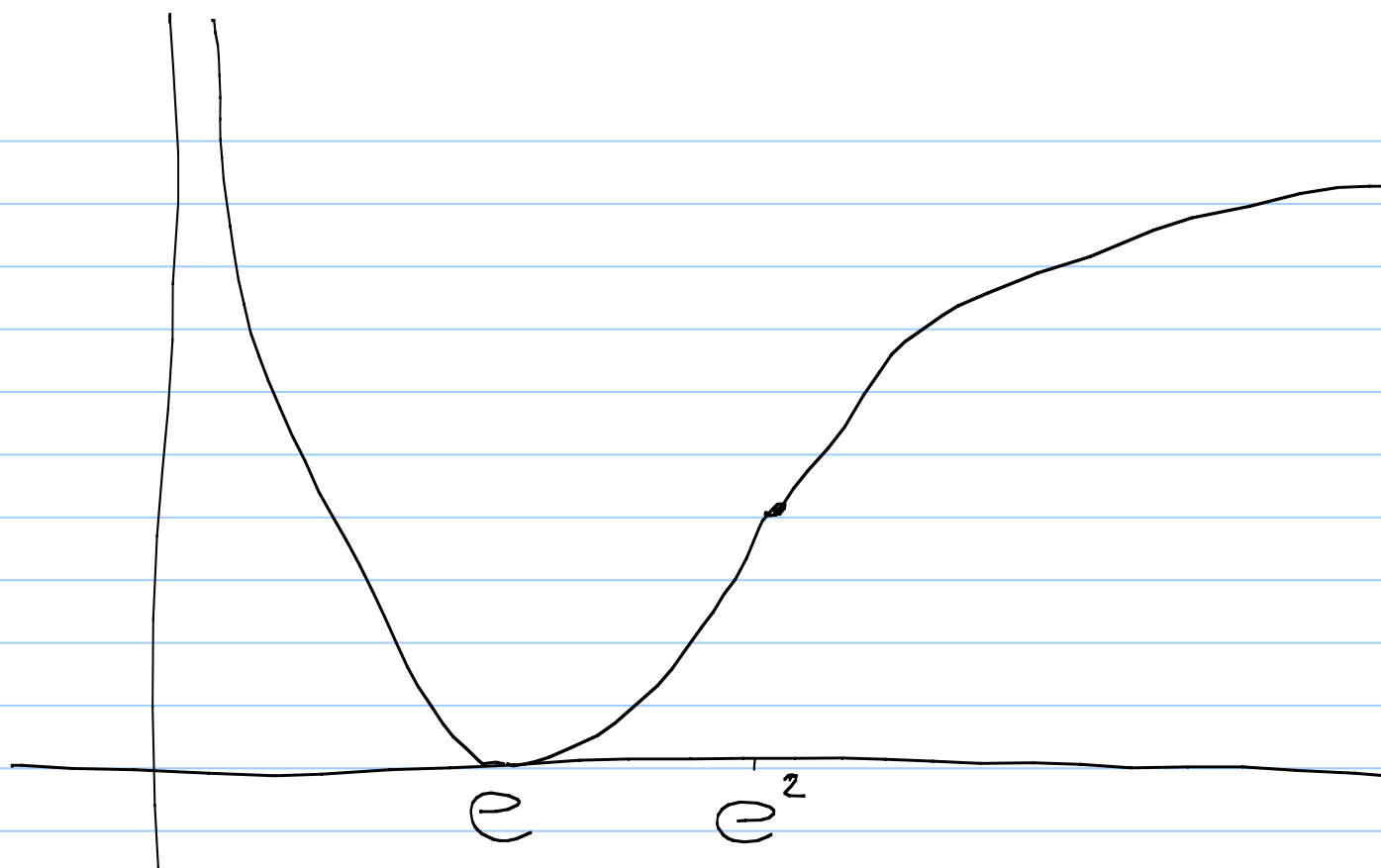


$\therefore f$ is concave up on $(0, e^2)$
and f is concave down on
 (e^2, ∞) .

g) Inflection point:

$$\begin{aligned} f(e^2) &= (1 - \ln e^2)^2 \\ &= (1 - 2)^2 \\ &= 1 \end{aligned}$$

$\therefore$ Inflection point at $(e^2, 1)$



Example

Sketch

1. continuous function $y = f(x)$
for $x \in (0, \infty)$.

2. $f(x) > 0 \quad \forall x \in (0, \infty)$

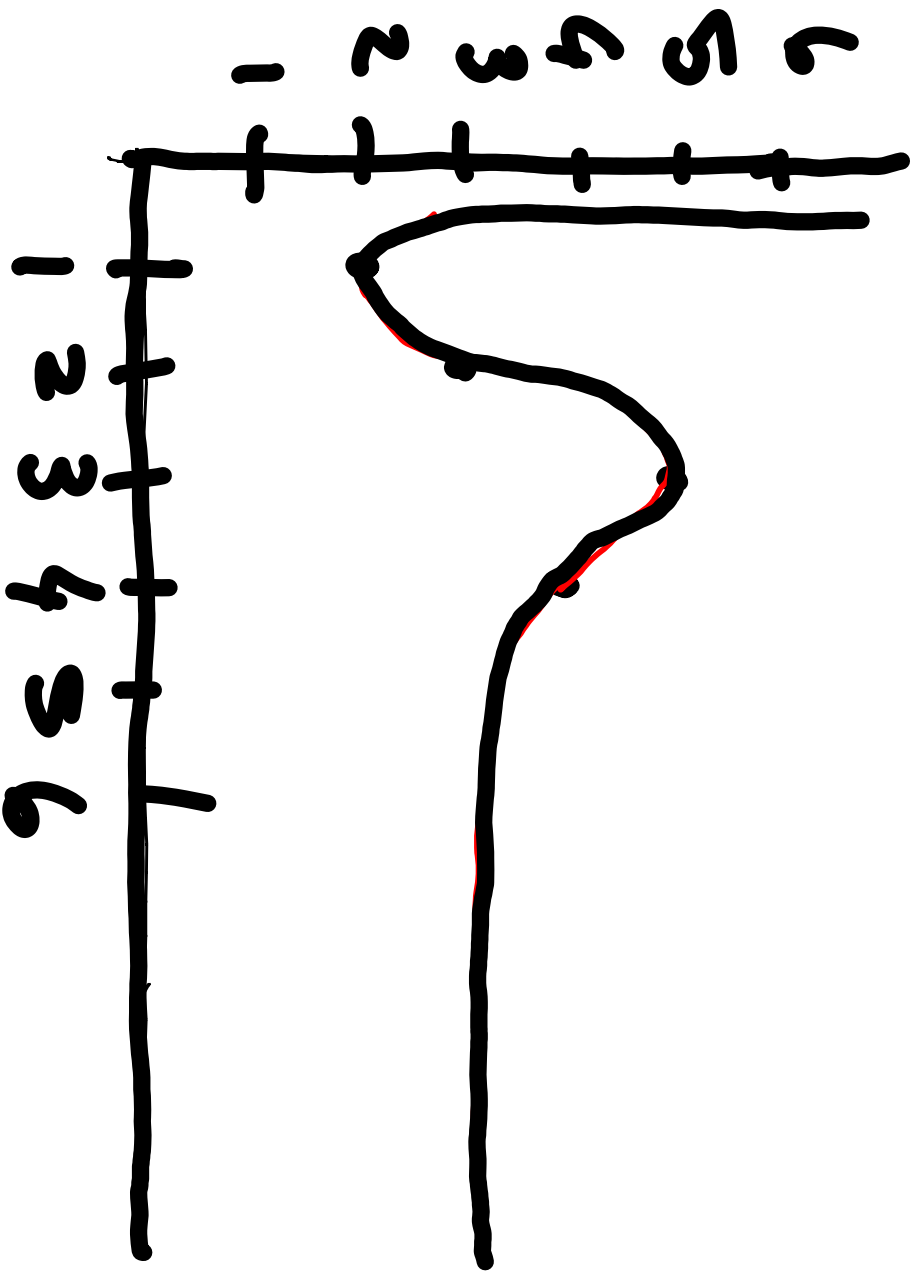
3. $x = 0$ is a vertical asymptote

4. $f(1) = 2$ is the global minimum.

5. $\lim_{x \rightarrow \infty} f(x) = 2$

6. f is increasing on $(1, 3)$
and decreasing on $(3, \infty)$

7. $f(2) = 2$ and $f(4) = 4$ are
points of inflection.



The function has a
local maximum at
 $x=3$.

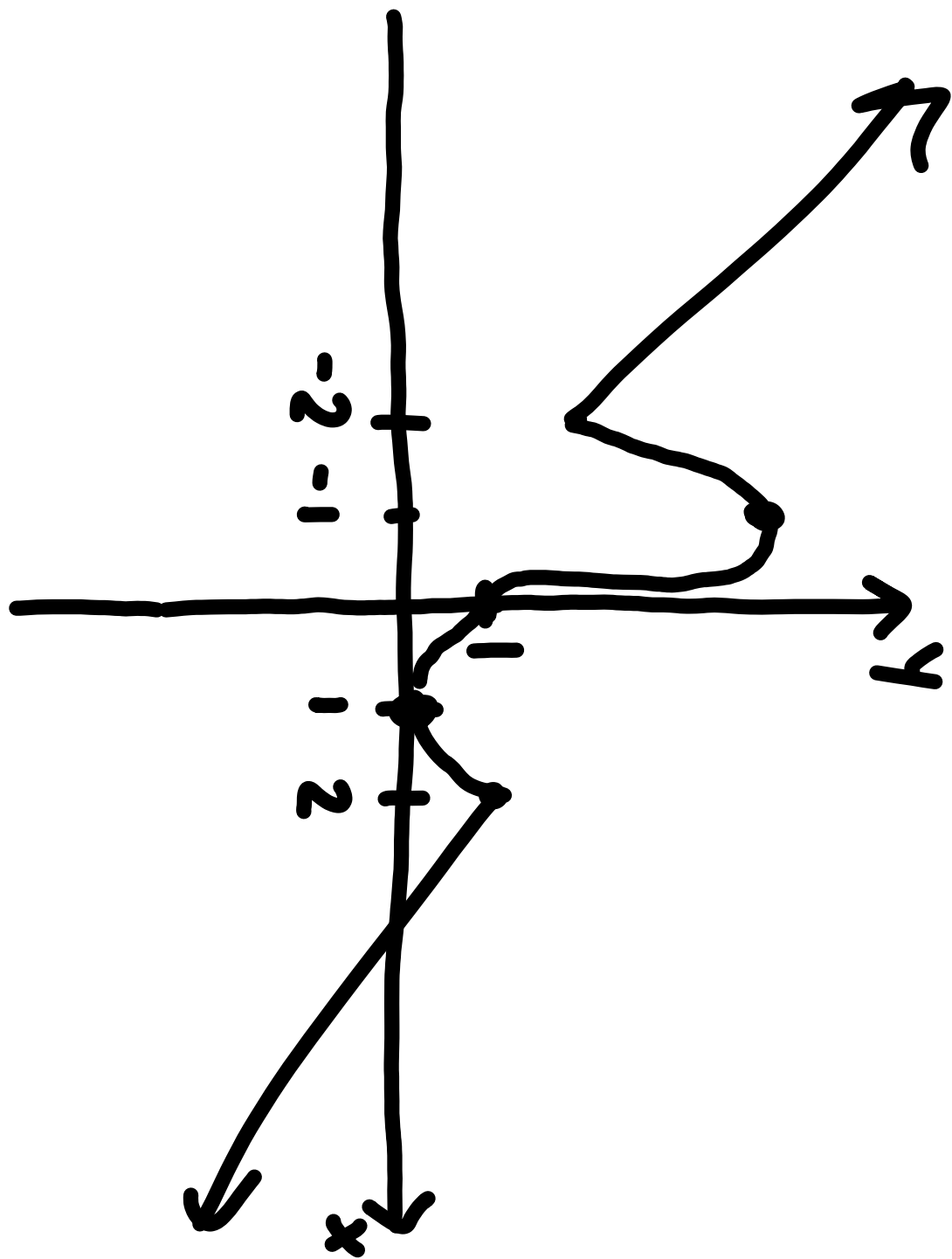
Example: Sketch a function that is continuous on $\mathbb{R}$ and satisfies the following conditions:

- $f'(-1) = f'(1) = 0$
- $f'(x) > 0$ if $x \in (-2, -1) \cup (1, 2)$
- $f'(x) < 0$ if $|x| < 1$

- $f'(x) = -1$ if $|x| > 2$

- $f''(x) < 0$ if $-2 < x < 0$

- Inflection point $(0, 1)$



Example

Sketch

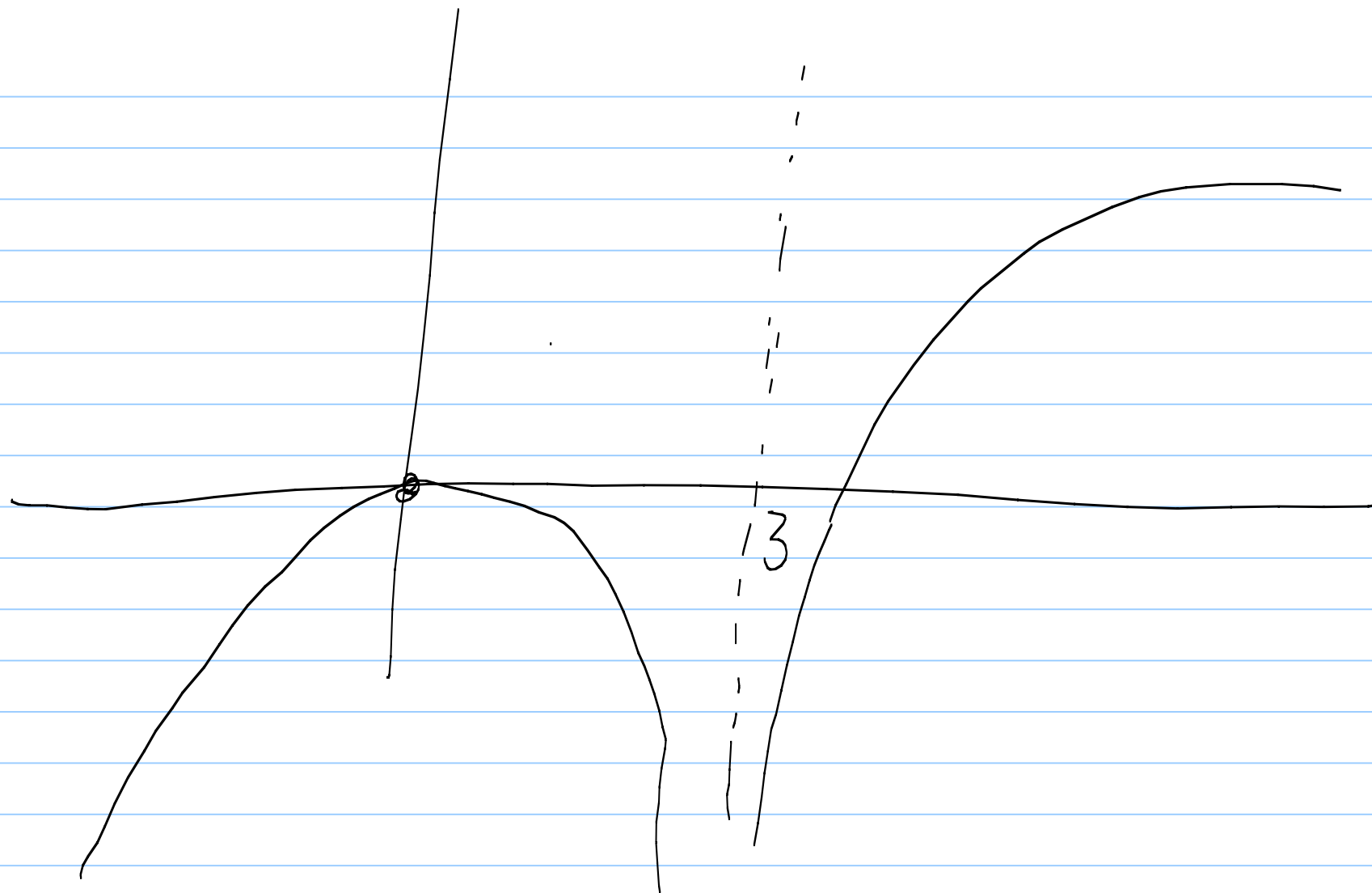
① $f'(0) = 0$

② $f'(x) > 0$ if $x < 0$ or $0 < x < 3$

③ $f'(x) < 0$ if $0 < x < 3$

④ $f''(x) < 0$ if $x \neq 3$

⑤ $\lim_{x \rightarrow 3} f(x) = -\infty$



Homework Sketch:

$$\textcircled{1} \quad f(x) = \frac{\ln x}{x^2}$$

$$\textcircled{2} \quad f(x) = x^2 e^{-x}$$

Solution to 1

$$y = \frac{\ln x}{x^2}$$

① Domain: $D_f = \{x \mid x > 0\}$
• $x \in (0, \infty)$.

② x -int, Let $y = 0$
 $\Rightarrow \ln x = 0$, $\Rightarrow x = 1$

No y-intercept

$$\textcircled{3} \quad f(-\infty) = \frac{\ln(-\infty)}{(-\infty)^2} = \frac{\ln(-x)}{x^2}$$

No symmetry.

\textcircled{4} Horizontal asymptote:

$$\lim_{\omega \rightarrow \infty} \frac{\ln \omega}{\omega^2} \quad \text{FORM.} \quad \left[\frac{\infty}{\infty} \right]$$

$$\stackrel{\text{L'Hop.}}{=} \lim_{\omega \rightarrow \infty} \frac{\frac{1}{\omega}}{2\omega}$$

$$= \lim_{\omega \rightarrow \infty} \frac{1}{2\omega^2} = 0$$

$\Rightarrow y=0$ is a H.O.A.

⑤ Vertical asymptotes

$$\lim_{x \rightarrow 0^+} \frac{\ln x}{x^2} = -\infty$$

$x=0$ is a vertical asym.

⑥. Increasing / decreasing.

$$f'(x) = \frac{\frac{1}{x} \cdot x^2 - \ln x \cdot 2x}{x^4}$$

$$= \frac{x [1 - 2 \ln x]}{x^4}$$

$$= \frac{1 - 2 \ln x}{x^3}$$

$$f'(x) = 0$$

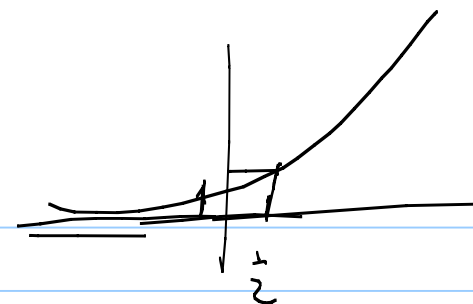
$$\frac{1 - 2 \ln x}{x^3} = 0.$$

$$1 - 2 \ln x = 0 \Rightarrow \ln x = \frac{1}{2}$$

$$\Rightarrow x = e^{1/2}.$$

... ect ...

y



Homework Find horizontal asymptotes
of $f(x) = x e^{-x^2/2}$